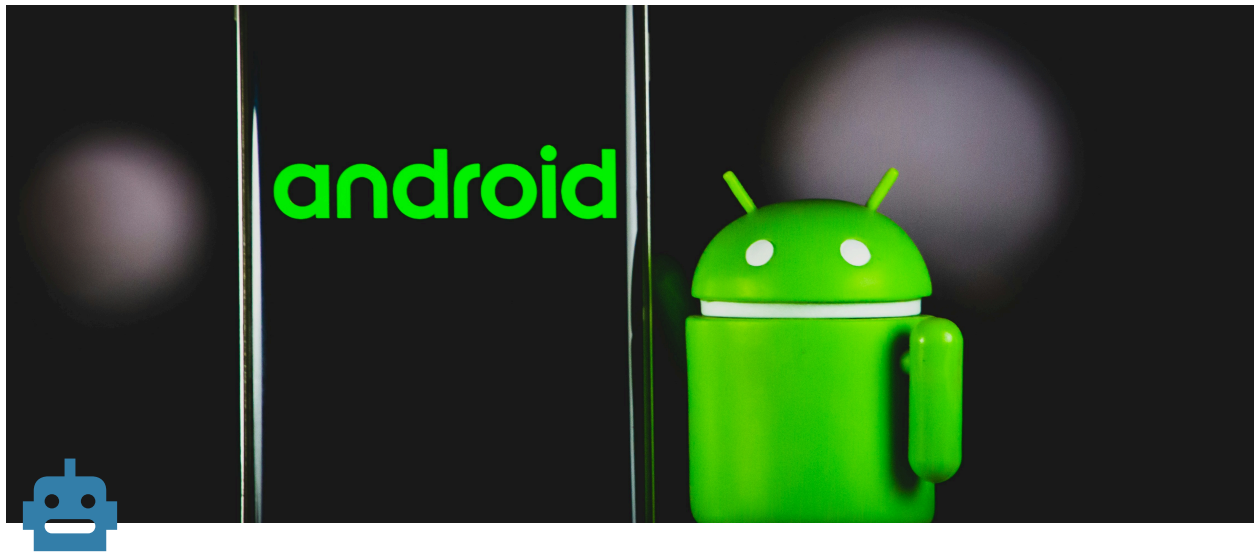




Android

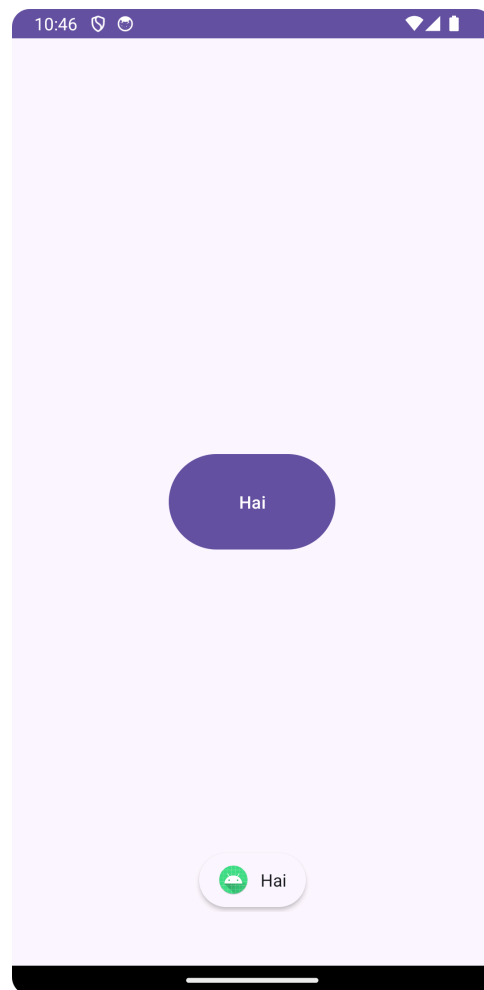
1 Implement Toast

```
<ConstraintLayout>
  <Button
    android:id="@+id/button"
    android:layout_width="136dp"
    android:layout_height="86dp"
    android:text="Hai"
    android:onClick="Show"
  />
</ConstraintLayout>
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```
public void Show(View v)
{
    Toast.makeText(this, "Hai", Toast.LENGTH_SHORT).show();
}
}
```



2.Implement a calculator

```
<ConstraintLayout>
  <EditText
    android:id="@+id/editTextNumber"
```

```
android:layout_width="356dp"
android:layout_height="65dp"
android:hint="No.1:"
android:inputType="number" />
```

<EditText

```
android:id="@+id/editTextNumber3"
android:layout_width="352dp"
android:layout_height="64dp"
android:hint="No.2:"
android:inputType="number" />
```

<Button

```
android:id="@+id/button"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="+"
android:onClick="function1" />
```

<Button

```
android:id="@+id/button2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="-"
android:onClick="function2" />
```

<Button

```
android:id="@+id/button3"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="X"
android:onClick="function3" />
```

<Button

```
android:id="@+id/button4"
android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="/"
        android:onClick="function4"/>
</ConstraintLayout>
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    public void function1(View v)
    {
        EditText txt1 = findViewById(R.id.editTextNumber);
        EditText txt2 = findViewById(R.id.editTextNumber3);
        int num1 = Integer.parseInt(txt1.getText().toString());
        int num2 = Integer.parseInt(txt2.getText().toString());
        int sum = num1 + num2;
        Toast.makeText(this, "Sum = " +sum, Toast.LENGTH_SHORT).show();
    }

    public void function2(View v)
    {
        EditText txt1 = findViewById(R.id.editTextNumber);
        EditText txt2 = findViewById(R.id.editTextNumber3);
        int num1 = Integer.parseInt(txt1.getText().toString());
        int num2 = Integer.parseInt(txt2.getText().toString());
        int Diff = num1 - num2;
        Toast.makeText(this, "Differnce = " +Diff, Toast.LENGTH_SHORT).show();
    }

    public void function3(View v)
    {
        EditText txt1 = findViewById(R.id.editTextNumber);
```

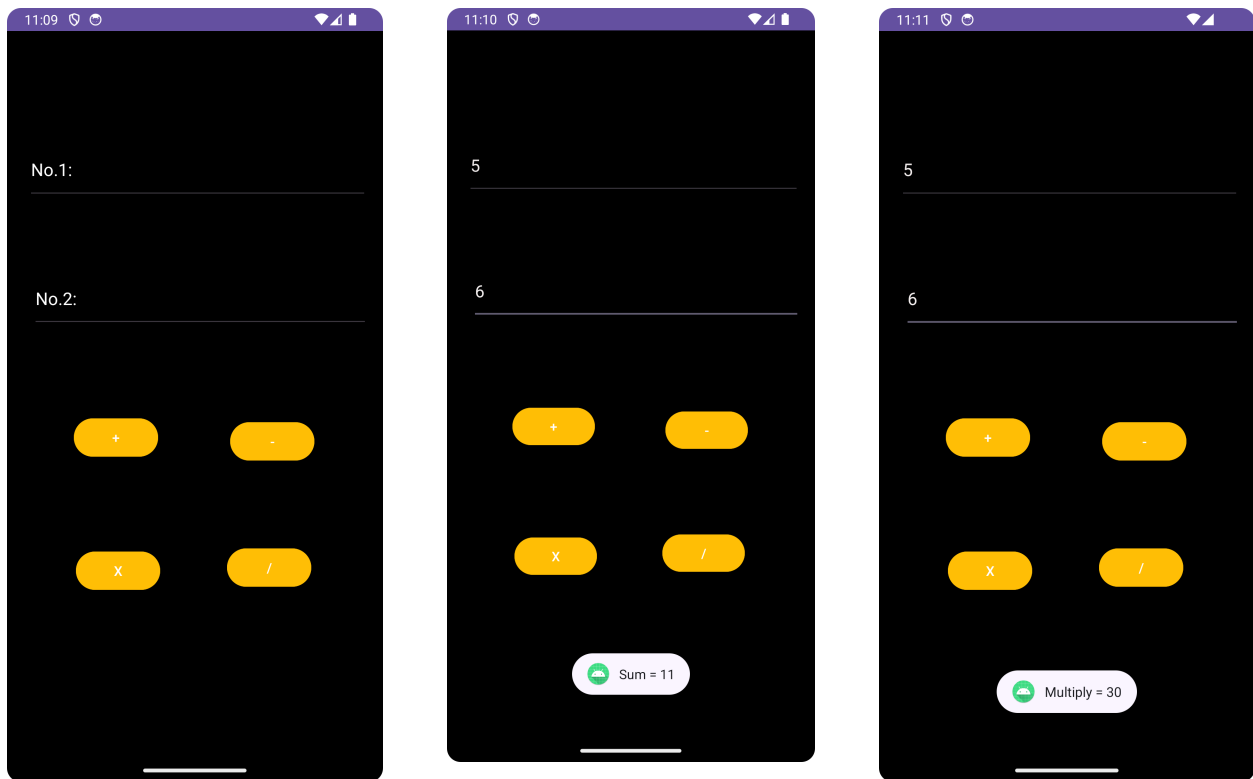


```

        EditText txt2 = findViewById(R.id.editTextNumber3);
        int num1 = Integer.parseInt(txt1.getText().toString());
        int num2 = Integer.parseInt(txt2.getText().toString());
        int mult = num1 * num2;
        Toast.makeText(this, "Multiply = " + mult, Toast.LENGTH_SHORT).show();
    }

    public void function4(View v)
    {
        EditText txt1 = findViewById(R.id.editTextNumber);
        EditText txt2 = findViewById(R.id.editTextNumber3);
        int num1 = Integer.parseInt(txt1.getText().toString());
        int num2 = Integer.parseInt(txt2.getText().toString());
        int Division = num1 / num2;
        Toast.makeText(this, "Division = " + Division, Toast.LENGTH_SHORT).show()
    }
}

```



3 Copy the content of EditText to TextView

<ConstraintLayout>

<EditText

```

    android:id="@+id/editTextNumber"
    android:layout_width="304dp"
    android:layout_height="58dp"
    android:ems="10"
    android:inputType="textPersonName"
    android:textAlignment="center" />

```

<Button

```

    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"

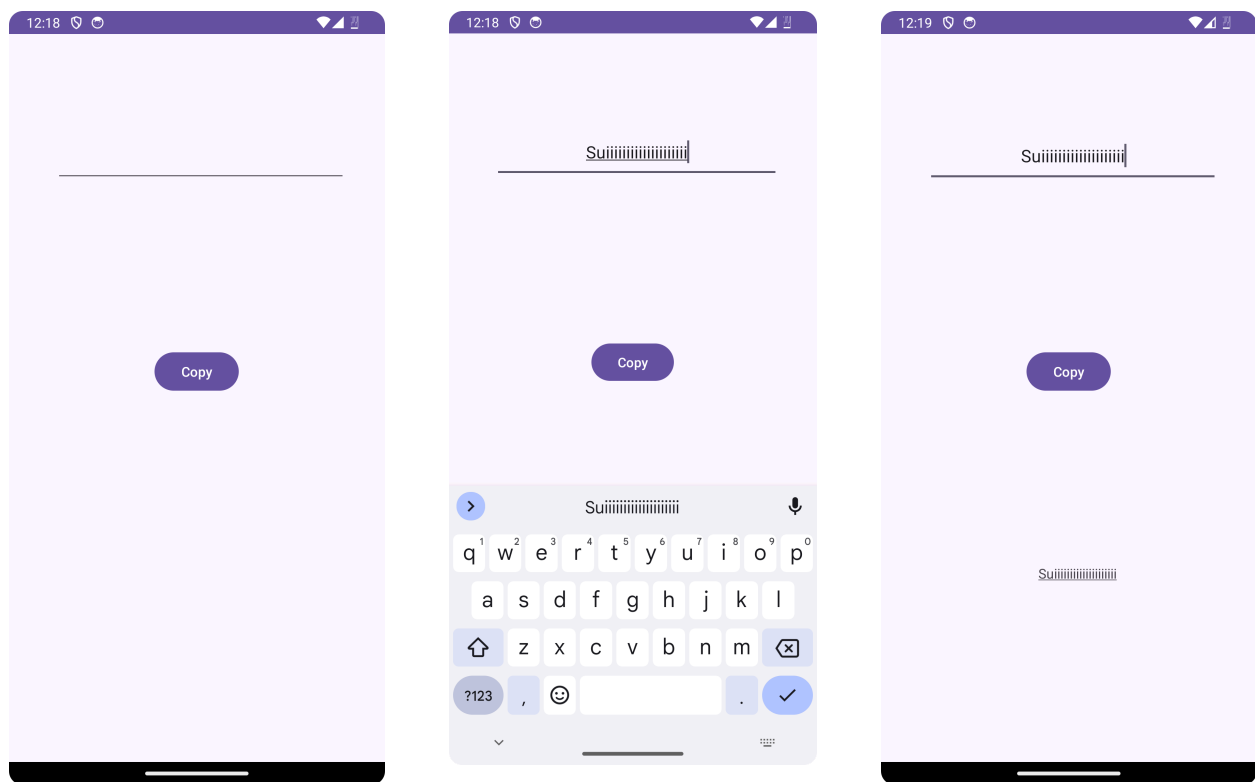
```

```
        android:text="Copy"
        android:onClick="copy" />

<TextView
    android:id="@+id/textView"
    android:layout_width="296dp"
    android:layout_height="60dp"
    android:textAlignment="center"/>
</ConstraintLayout>
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void copy(View v)
    {
        EditText e1 = findViewById(R.id.editTextNumber);
        TextView tv = findViewById(R.id.textView);
        tv.setText(e1.getText());
    }
}
```



4 Navigate from one activity to another

activity_main.xml

```
<ConstraintLayout>

    <Button
        android:id="@+id/button"
        android:layout_width="166dp"
        android:layout_height="70dp"
        android:text="Next"
        android:textSize="24sp"
        android:onClick="Next"/>

</ConstraintLayout>
```

activity_second.xml

```

<ConstraintLayout>

    <TextView
        android:id="@+id/textView"
        android:layout_width="142dp"
        android:layout_height="88dp"
        android:text="Haiiiiiiii"
        android:textAlignment="center"
        android:textSize="34sp"
        android:textStyle="bold" />

</ConstraintLayout>

```

Activity_Main.java

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void Next(View v){
        Intent i = new Intent(this, MainActivity2.class);
        startActivity(i);
    }
}

```

SecondActivity.java

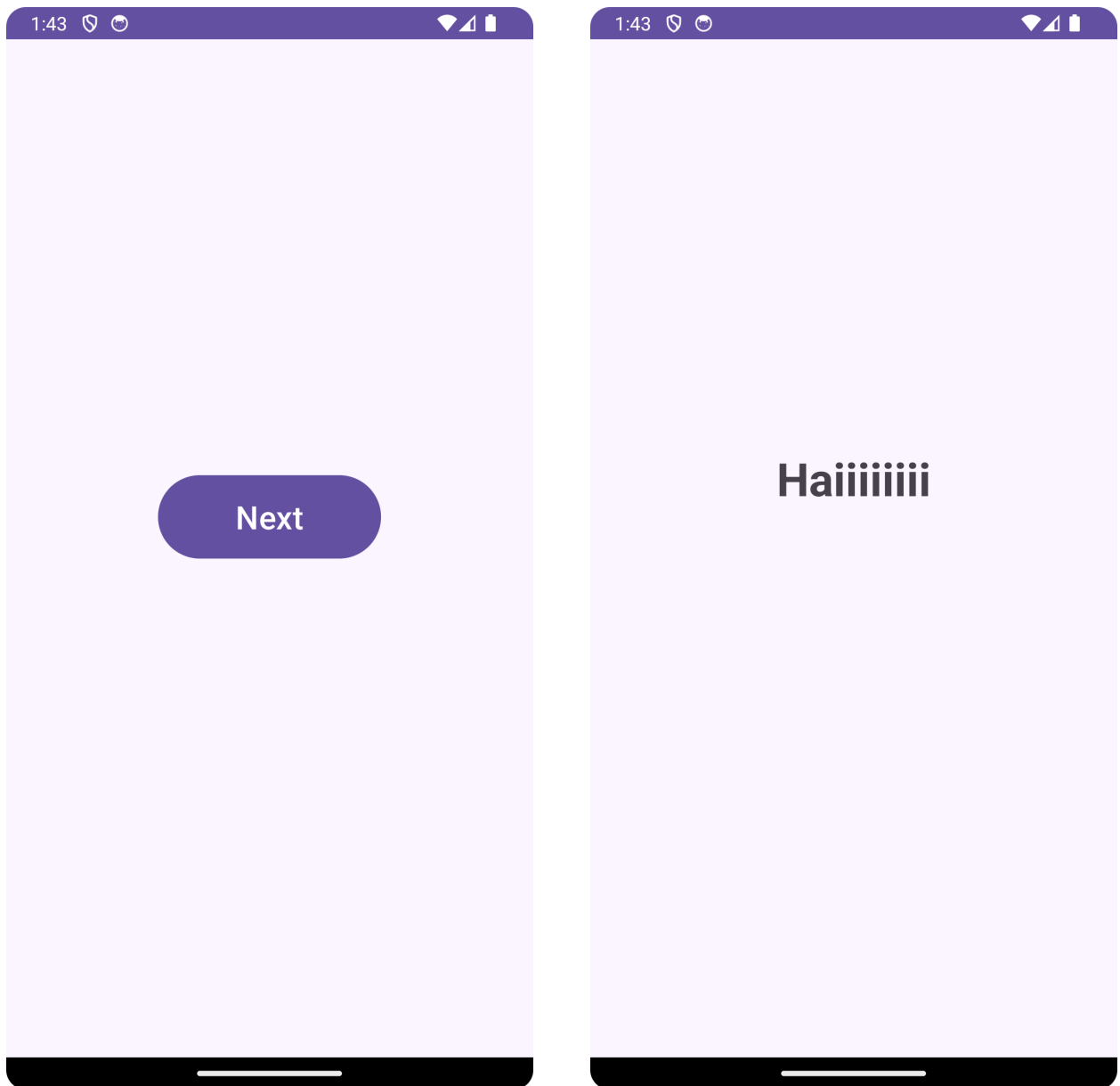
```

public class MainActivity2 extends AppCompatActivity {

    @Override

```

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_second);  
}  
}
```



5.Implement ImageView

<ConstraintLayout

<ImageView

```
    android:id="@+id/imageView"  
    android:layout_width="215dp"  
    android:layout_height="149dp"/>
```

<Button

```
    android:id="@+id/button5"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Rose"/>
```

<Button

```
    android:id="@+id/button6"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Lily" />
```

<Button

```
    android:id="@+id/button7"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Lotus" />
```

<Button

```
    android:id="@+id/button8"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Tulip"/>
```

</ConstraintLayout>

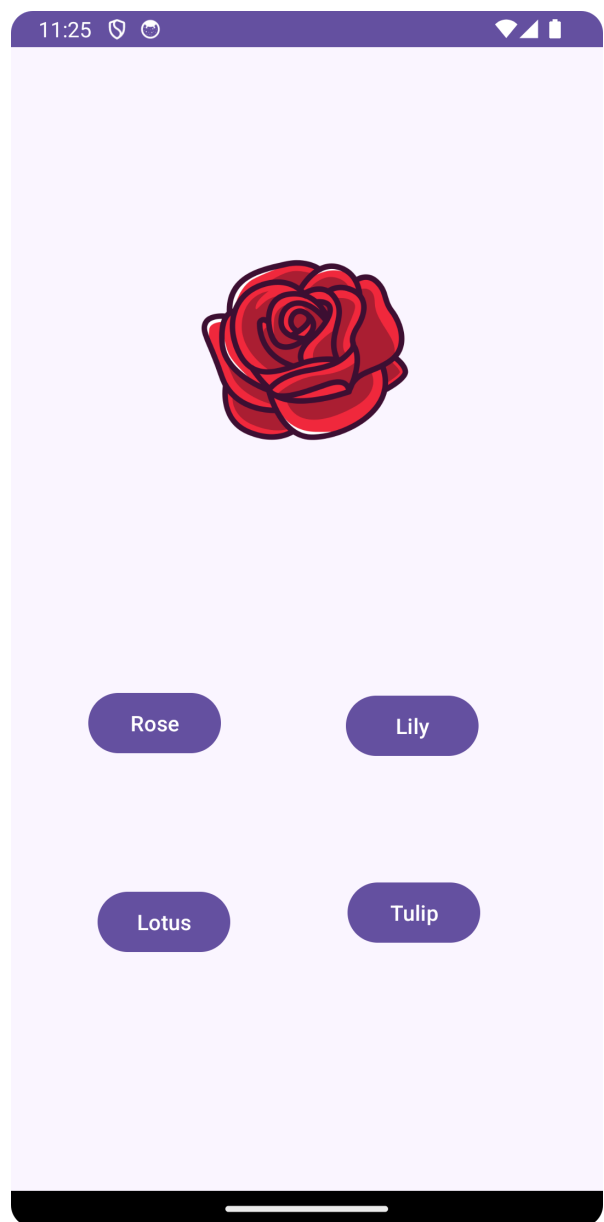
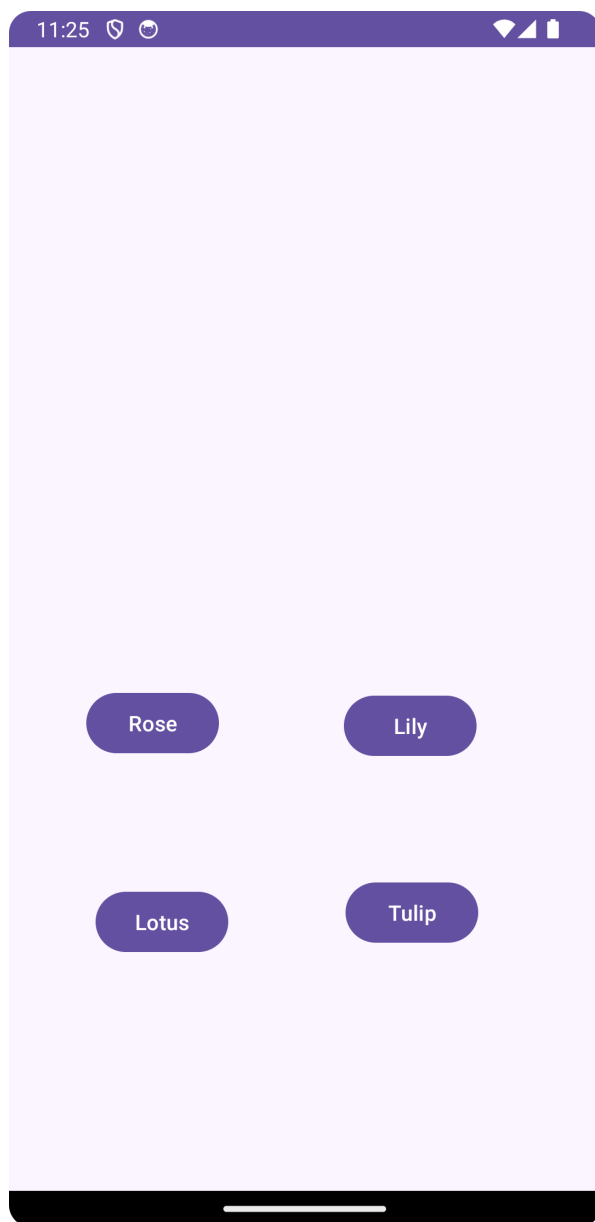
```

public class MainActivity extends AppCompatActivity implements View.OnClickListener {
    ImageView iv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        iv = findViewById(R.id.imageView);
        Button b1 = findViewById(R.id.button5);
        Button b2 = findViewById(R.id.button6);
        Button b3 = findViewById(R.id.button7);
        Button b4 = findViewById(R.id.button8);
        b1.setOnClickListener(this);
        b2.setOnClickListener(this);
        b3.setOnClickListener(this);
        b4.setOnClickListener(this);
    }

    @Override
    public void onClick(View v) {
        if(v.getId()==R.id.button5)
        {
            iv.setImageResource(R.drawable.rose);
        }
        else if (v.getId()==R.id.button6) {
            iv.setImageResource(R.drawable.lilly);
        }
        else if (v.getId()==R.id.button7) {
            iv.setImageResource(R.drawable.lotus);
        }
        else {
            iv.setImageResource(R.drawable.tulip);
        }
    }
}

```

6 Implement Spinner

String.xml

```
<resources>
  <string name="app_name">Spinner</string>
  <string-array name="Flower">
    <item>Lily</item>
```

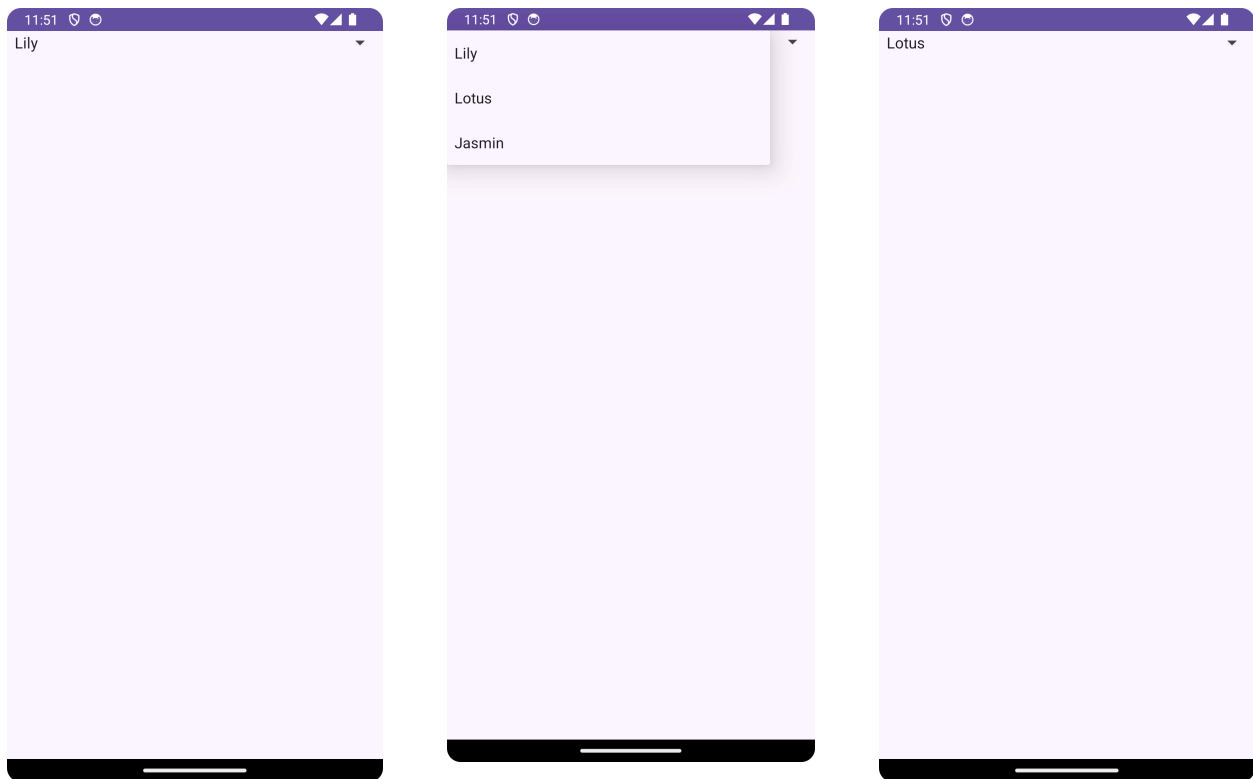
```
<item>Lotus</item>
<item>Jasmin</item>
</string-array>
</resources>
```

```
<LinearLayout
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:entries="@array/Flower"/>
</LinearLayout>
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.layout);
    }
}
```



7.Implement List

```
<LinearLayout
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ListView
        android:id="@+id/l1"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

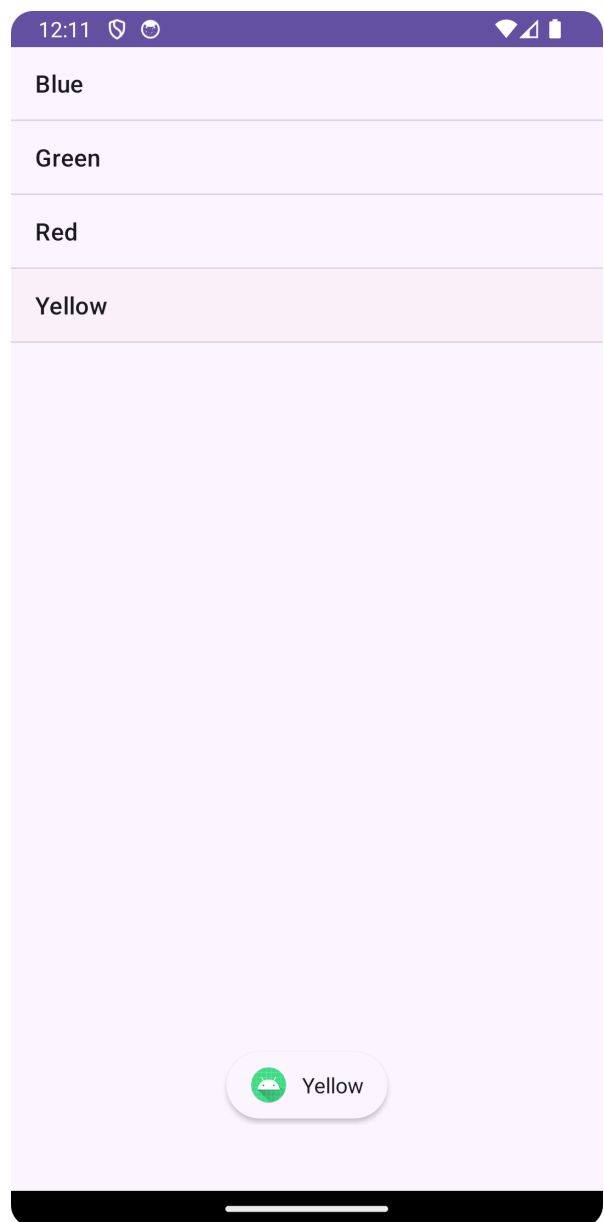
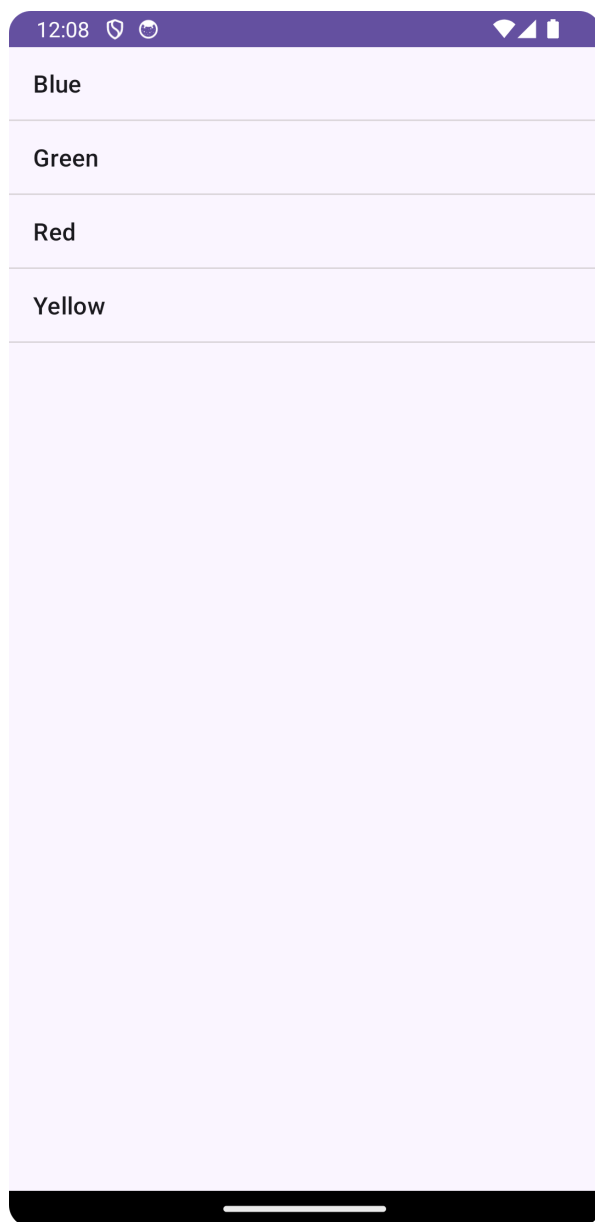
```
public class MainActivity extends AppCompatActivity implements AdapterView.C
    @Override
```

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.linear);
    ListView lv = findViewById(R.id.l1);
    String f1[]={"Blue","Green","Red","Yellow"};
    ArrayAdapter<String> ad = new ArrayAdapter<String>(this, android.R.layout
    lv.setAdapter(ad);
    lv.setOnItemClickListener(this);
}

@Override
public void onItemClick(AdapterView<?> parent, View view, int position, long id) {
    String s = parent.getItemAtPosition(position).toString();
    Toast.makeText(this, s, Toast.LENGTH_SHORT).show();
}
}

```



8.Implement Adaptor

```
<ConstraintLayout>
  <ListView
    android:id="@+id/lv"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:layout_editor_absoluteX="1dp"
```

```
tools:layout_editor_absoluteY="1dp"  
/>  
</ConstraintLayout>
```

```
public class MainActivity extends AppCompatActivity implements AdapterView.C
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
    String f1[]={"Rose","Lily","Lotus"};
```

```
    ArrayAdapter <String> ad = new ArrayAdapter<String>(this, android.R.layout
```

```
    ListView lv =findViewById(R.id.lv);
```

```
    lv.setAdapter(ad);
```

```
    lv.setOnItemClickListener(this);
```

```
}
```

```
@Override
```

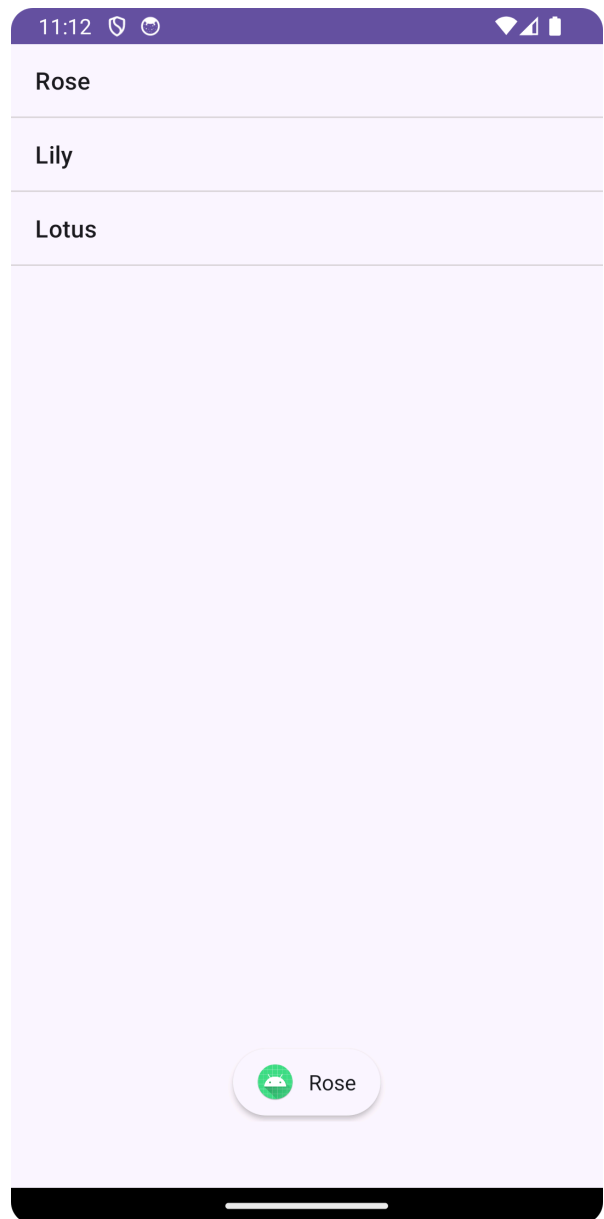
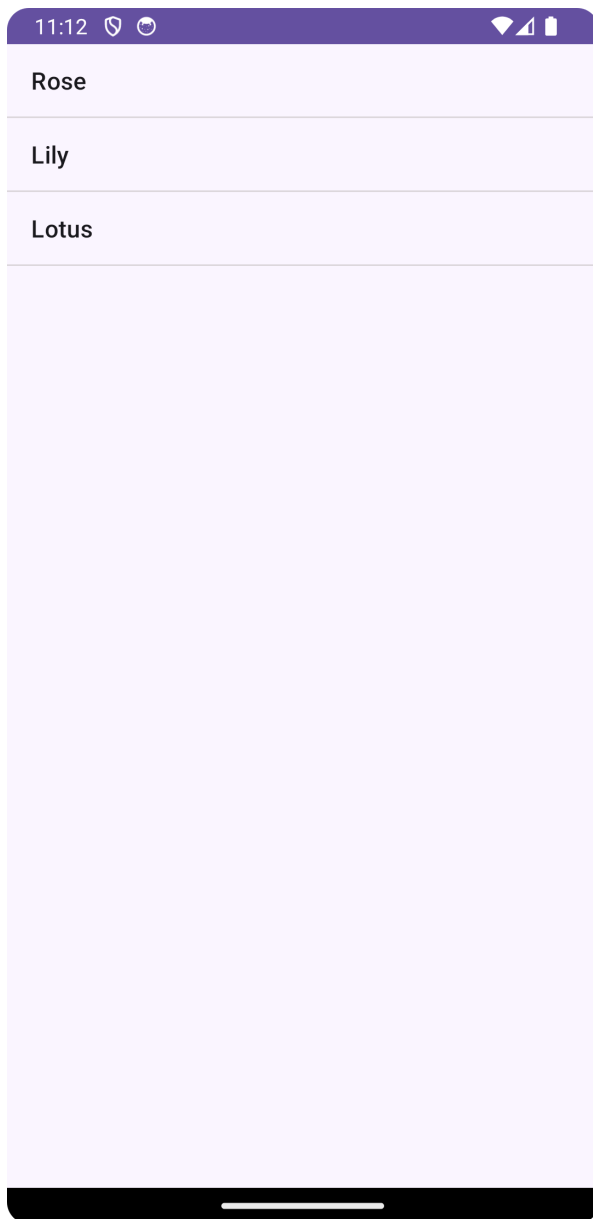
```
public void onItemClick(AdapterView<?> parent, View view, int position, long i
```

```
    String s = parent.getItemAtPosition(position).toString();
```

```
    Toast.makeText(this, s, Toast.LENGTH_SHORT).show();
```

```
}
```

```
}
```



9 Implement GridView

```
<ConstraintLayout>
```

```
<GridView
```

```
    android:id="@+id/gv"
```

```
    android:layout_width="409dp"
```

```
    android:layout_height="729dp"
```

```
        android:numColumns="3"/>
    </ConstraintLayout>
```

```
public class MainActivity extends AppCompatActivity implements AdapterView.C
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        GridView gv = findViewById(R.id.gv);
```

```
        String[] a = {"A","B","C","D","E","F"};
```

```
        ArrayAdapter<String> ad = new ArrayAdapter<String>(this, android.R.layout
```

```
        gv.setAdapter(ad);
```

```
        gv.setOnItemClickListener(this);
```

```
    }
```

```
    @Override
```

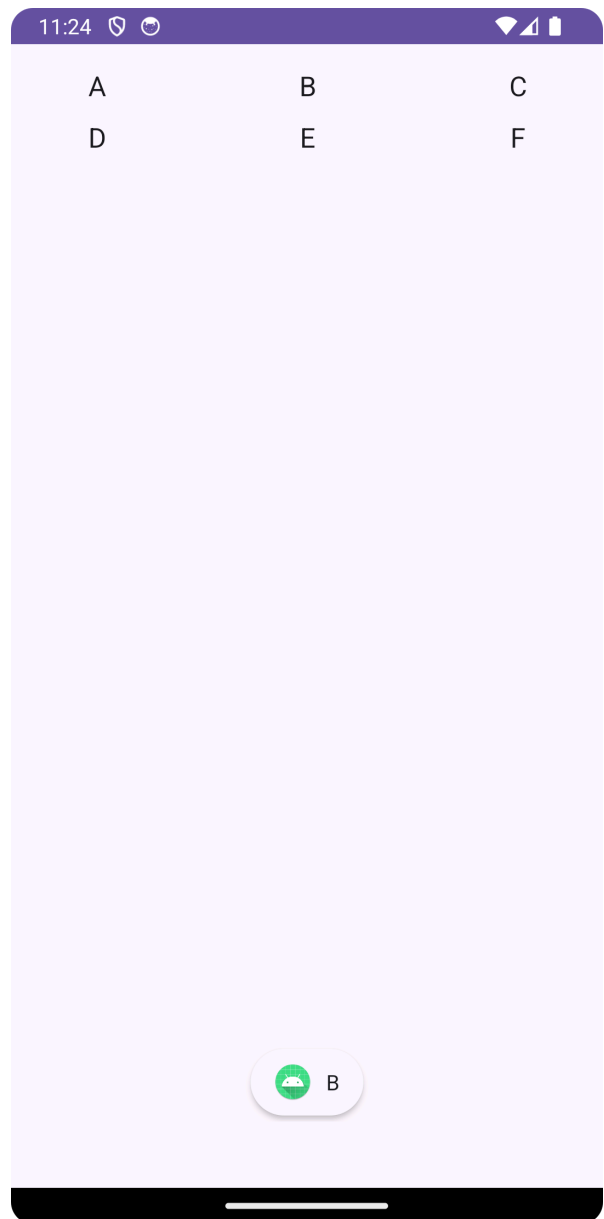
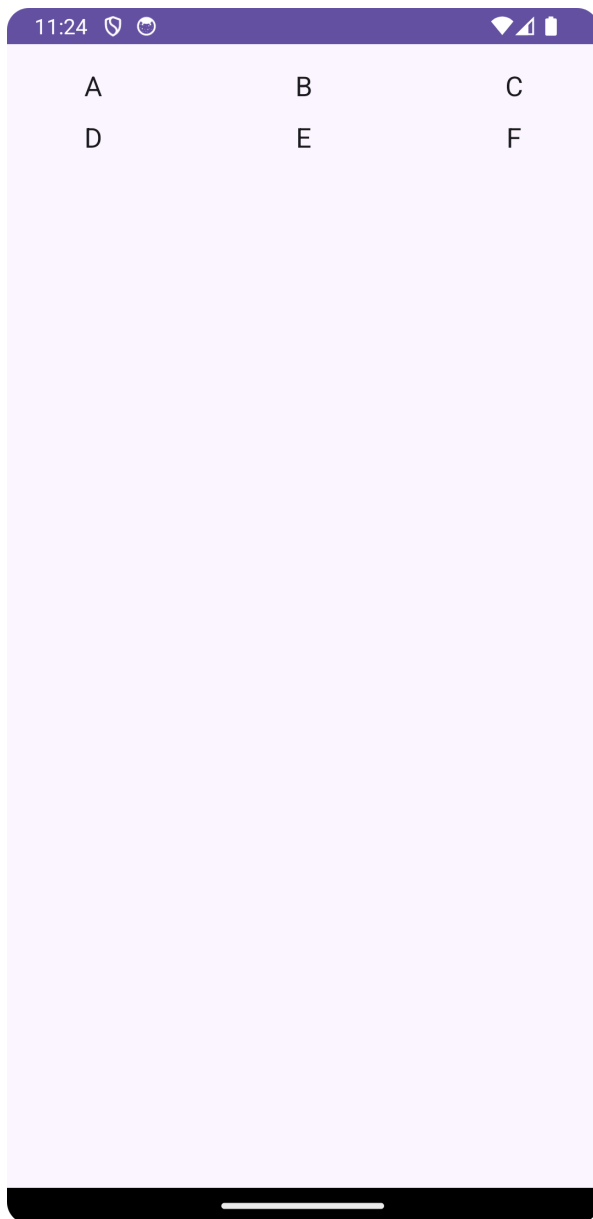
```
    public void onItemClick(AdapterView<?> parent, View view, int position, long i
```

```
        String s = parent.getItemAtPosition(position).toString();
```

```
        Toast.makeText(this, s, Toast.LENGTH_SHORT).show();
```

```
    }
```

```
}
```

10 Implement Menu

```
<menu>

  <item
    android:id="@+id/m1"
    android:title="Settings" />
  <item
```

```

        android:id="@+id/m2"
        android:title="About" />
    <item
        android:id="@+id/m3"
        android:title="Version" />
</menu>

```

```

public class MainActivity extends AppCompatActivity {

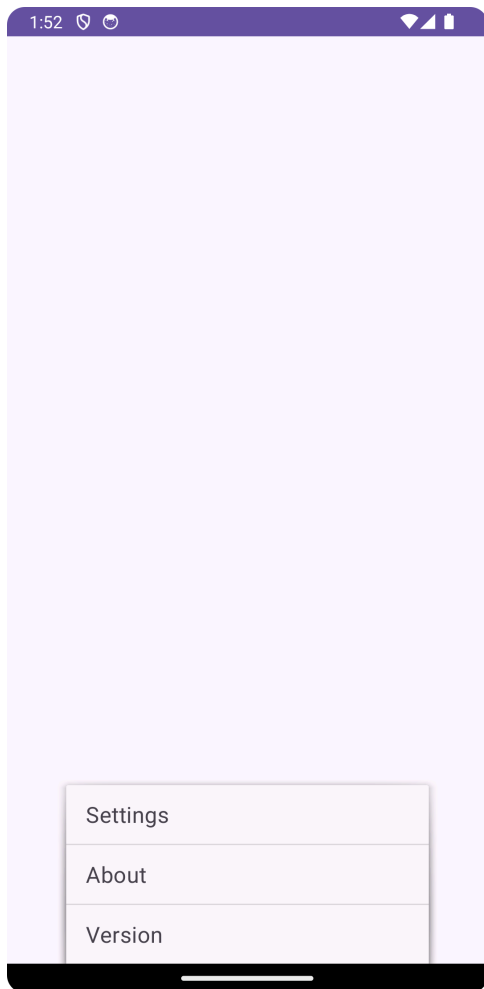
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        MenuInflater i = getMenuInflater();
        i.inflate(R.menu.menu, menu);
        return super.onCreateOptionsMenu(menu);
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.m1)
        {
            Toast.makeText(this, "You selected Settings", Toast.LENGTH_SHORT).show()
        }
        else if (id == R.id.m2)
        {
            Toast.makeText(this, "You selected About", Toast.LENGTH_SHORT).show()
        }
        else {
            Toast.makeText(this, "You selected Versions", Toast.LENGTH_SHORT).show()
        }
    }
}

```

```
        return super.onOptionsItemSelected(item);  
    }  
}
```



11 Implement CheckBox

```
<ConstraintLayout>
```

```
    <CheckBox
```

```
        android:id="@+id/checkBox"
```

```
        android:layout_width="208dp"
```

```
        android:layout_height="54dp"
```

```
        android:text="PG"/>
```

```

<CheckBox
    android:id="@+id/checkBox2"
    android:layout_width="211dp"
    android:layout_height="54dp"
    android:text="UG"/>

<CheckBox
    android:id="@+id/checkBox3"
    android:layout_width="213dp"
    android:layout_height="62dp"
    android:text="XII"/>

<CheckBox
    android:id="@+id/checkBox4"
    android:layout_width="198dp"
    android:layout_height="62dp"
    android:text="X" />

<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click"
    android:onClick="show" />
</ConstraintLayout>

```

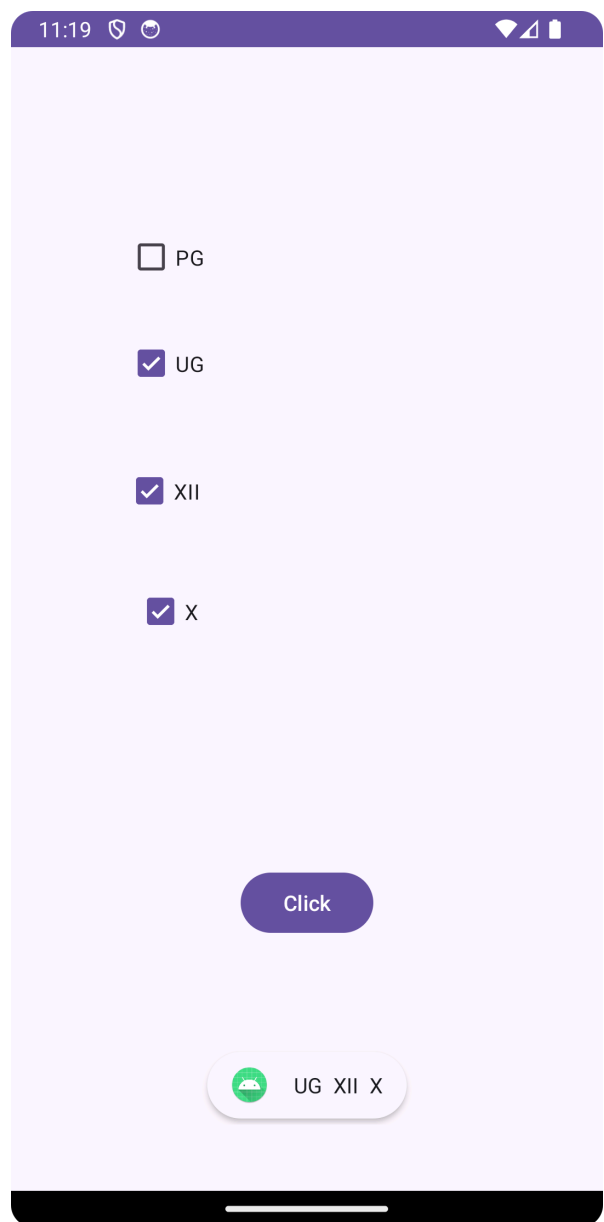
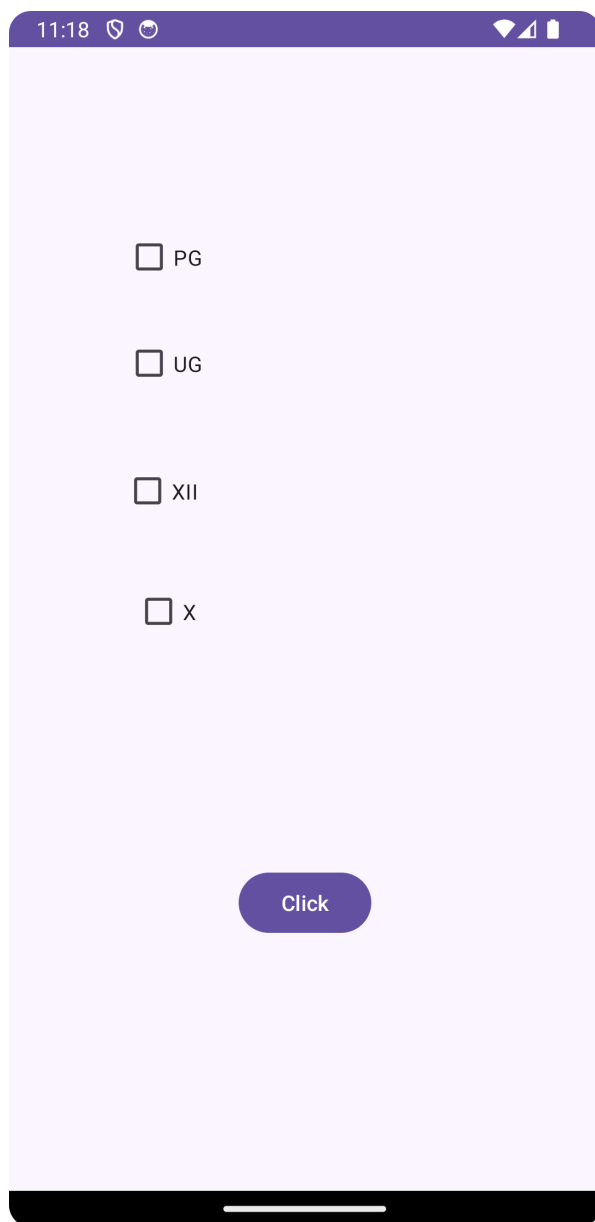
```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void show(View v)

```

```
{
    String msg = " ";
    CheckBox c1 = findViewById(R.id.checkBox);
    CheckBox c2 = findViewById(R.id.checkBox2);
    CheckBox c3 = findViewById(R.id.checkBox3);
    CheckBox c4 = findViewById(R.id.checkBox4);
    if(c1.isChecked())
        msg += " PG ";
    if(c2.isChecked())
        msg += " UG ";
    if(c3.isChecked())
        msg += " XII ";
    if(c4.isChecked())
        msg += " X";
    Toast.makeText(this, msg, Toast.LENGTH_SHORT).show();
}
}
```



12 Implement RadioButton

```
<ConstraintLayout>
```

```
<RadioGroup
```

```
    android:id="@+id/radiogroup"
```

```
    android:layout_width="264dp"
```

```
    android:layout_height="284dp" >
```

```

<RadioButton
    android:id="@+id/radioButton2"
    android:layout_width="match_parent"
    android:layout_height="90dp"
    android:text="Male" />

<RadioButton
    android:id="@+id/radioButton3"
    android:layout_width="match_parent"
    android:layout_height="96dp"
    android:text="Female" />

<RadioButton
    android:id="@+id/radioButton4"
    android:layout_width="match_parent"
    android:layout_height="100dp"
    android:text="Other" />
</RadioGroup>
</ConstraintLayout>

```

public class MainActivity extends AppCompatActivity implements RadioGroup.OnCheckedChangeListener {

 @Override

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    RadioGroup rg = findViewById(R.id.radiogroup);
    rg.setOnCheckedChangeListener(this);
}

```

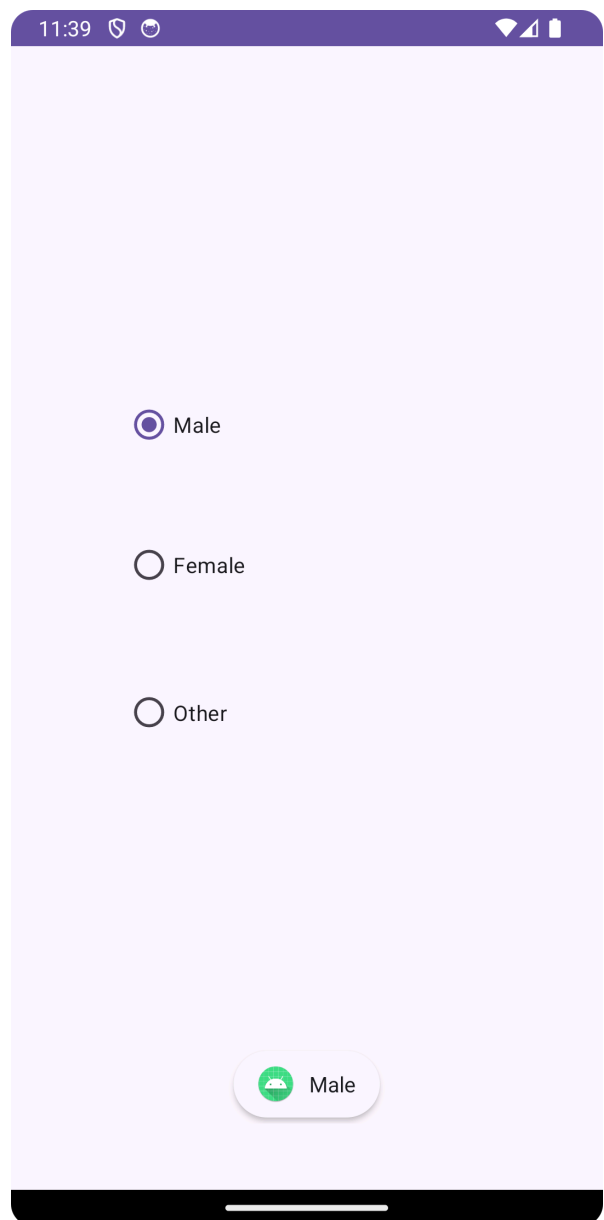
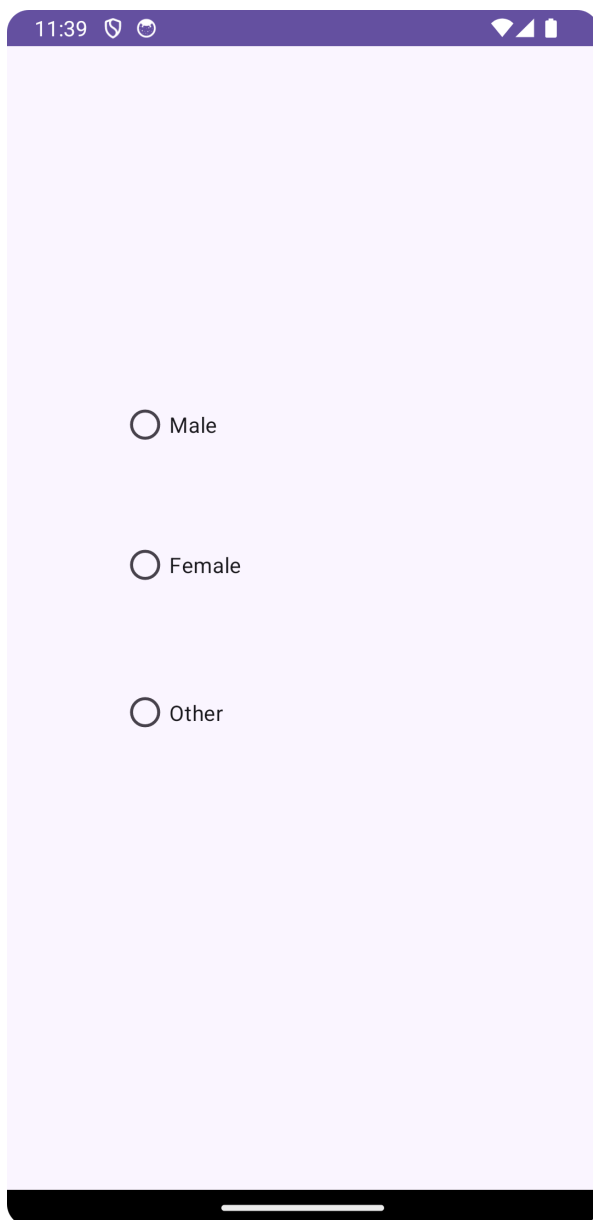
 @Override

```

public void onCheckedChanged(RadioGroup group, int checkedId) {
    String msg = "";
    if (checkedId == R.id.radioButton2)

```

```
        msg = "Male";  
    else if (checkedId == R.id.radioButton3)  
        msg = "Female";  
    else  
        msg = "Other";  
    Toast.makeText(this,msg, Toast.LENGTH_SHORT).show();  
}  
}
```



13 Implement Date Picker and Time Picker

```
<LinearLayout>

    <TimePicker
        android:id="@+id/tp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <DatePicker
        android:id="@+id/dp"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

public class MainActivity extends AppCompatActivity implements TimePicker.On

@Override

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.linear);
    TimePicker tp = findViewById(R.id.tp);
    DatePicker dp = findViewById(R.id.dp);
    tp.setOnTimeChangedListener(this);
}
```

@Override

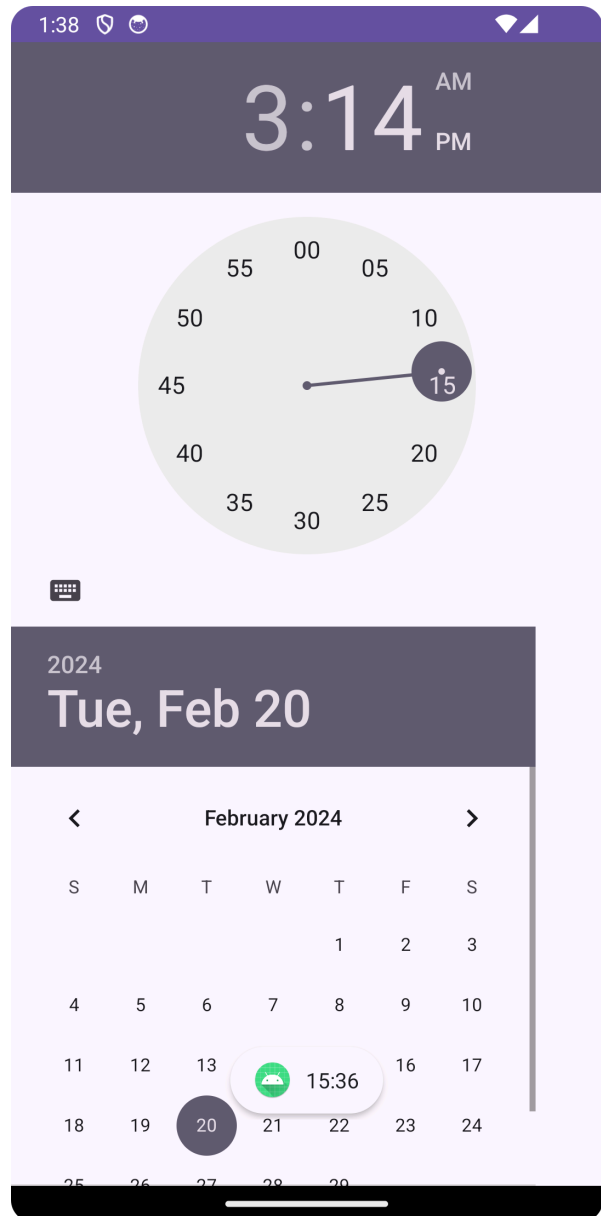
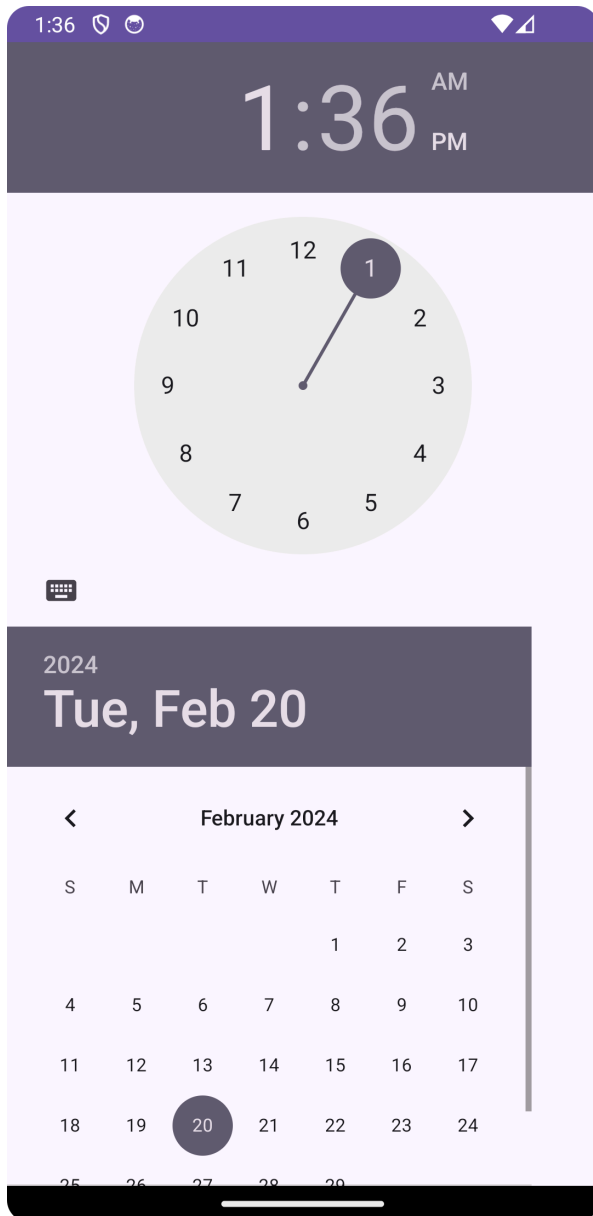
```
public void onChanged(DatePicker view, int year, int monthOfYear, int day) {
    String dp = year + "/" + (monthOfYear+1) + "/" + dayOfMonth;
    Toast.makeText(this, dp, Toast.LENGTH_SHORT).show();
}
```

@Override

```

public void onTimeChanged(TimePicker view, int hourOfDay, int minute) {
    String tp = hourOfDay+": "+minute;
    Toast.makeText(this, tp, Toast.LENGTH_SHORT).show();
}
}

```



14 Implement Alert

```

<ConstraintLayout>

    <Button
        android:id="@+id/button"
        android:layout_width="192dp"
        android:layout_height="82dp"
        android:text="Exit"
        android:onClick="show"
        android:textSize="34sp"/>
</ConstraintLayout>

```

```

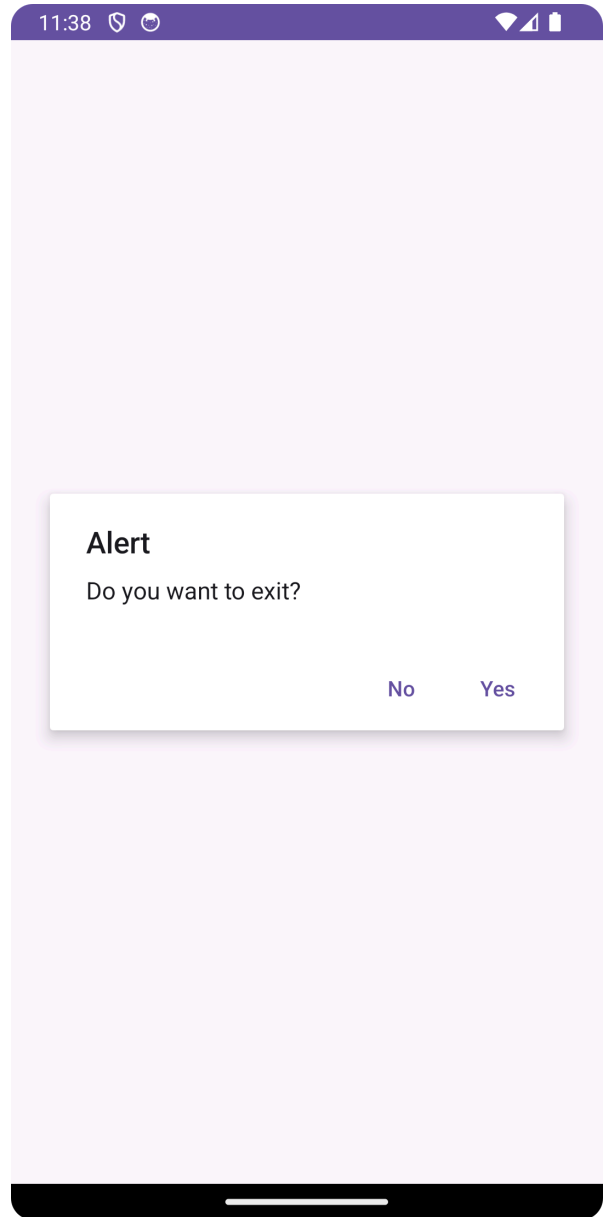
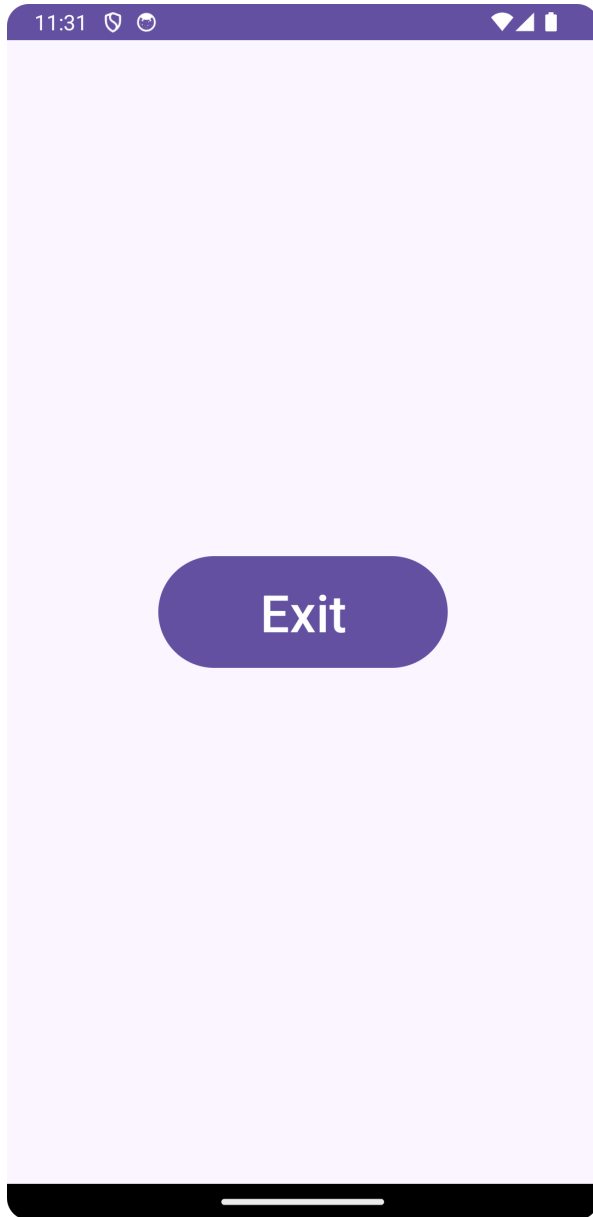
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    public void show(View v)
    {
        AlertDialog.Builder b = new AlertDialog.Builder(this);
        b.setTitle("Alert");
        b.setMessage("Do you want to exit?");
        b.setPositiveButton("Yes", new DialogInterface.OnClickListener()
        {
            public void onClick(DialogInterface dialog, int which)
            {
                finish();
            }
        });
        b.setNegativeButton("No",null);
        AlertDialog ad = b.create();
        ad.show();
    }
}

```

```
}  
}
```



15 Implement Date Picker Dialogue and Time Picker Dialogue

```
<LinearLayout>
```

```
<TextView
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/textview" />
```

<Button

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/b1"
    android:text="Set Time"
    android:onClick="Set" />
```

<TextView

```
    android:id="@+id/textview2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text=" " />
```

<Button

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/b2"
    android:text="Set Date"
    android:onClick="Set" />
```

</LinearLayout>

public class MainActivity extends AppCompatActivity implements DatePickerDialog

```
    TextView tv1;
    TextView tv2;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.linear);
        tv1 = findViewById(R.id.textview);
        tv2 = findViewById(R.id.textview2);
```

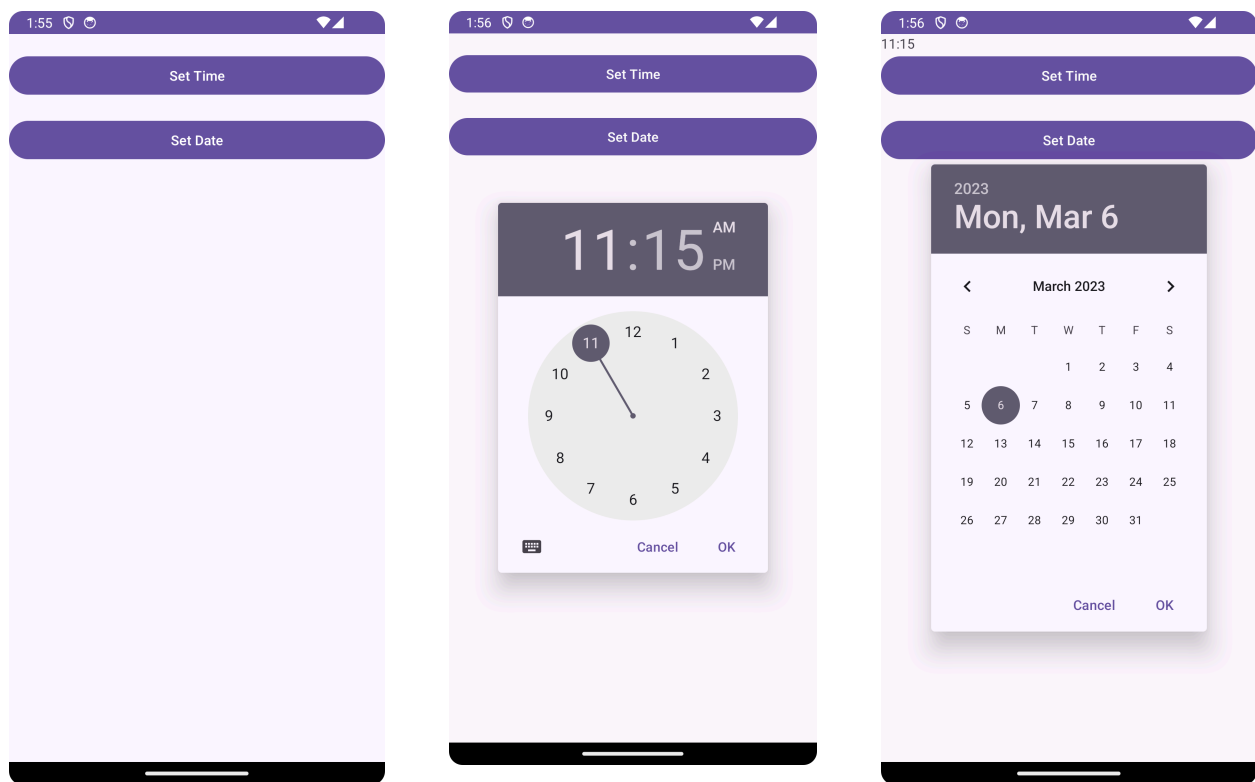
```

    }
    public void Set(View v){
        if(v.getId()==R.id.b1){
            TimePickerDialog tp = new TimePickerDialog(this,this,11,15,false);
            tp.show();
        }
        if (v.getId()==R.id.b2){
            DatePickerDialog dp = new DatePickerDialog(this,this,2023,2,6);
            dp.show();
        }
    }

    @Override
    public void onDateSet(DatePicker view, int year, int month, int dayOfMonth) {
        tv2.setText(year+"/"+(month+1)+"/"+dayOfMonth);
    }

    @Override
    public void onTimeSet(TimePicker view, int hourOfDay, int minute) {
        tv1.setText(hourOfDay+": "+minute);
    }
}

```



16 Implement Chronometer

<ConstraintLayout>

<Chronometer

```
android:id="@+id/c"
android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
```

<TextView

```
android:id="@+id/textView"
android:layout_width="373dp"
android:layout_height="140dp"
android:textAlignment="center"
android:textSize="48sp"
android:text="00:00:00"/>
```

```
<Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Start"
    android:onClick="Start"/>
```

```
<Button
    android:id="@+id/button4"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Pause"
    android:onClick="Pause" />
```

```
<Button
    android:id="@+id/button5"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Reset"
    android:onClick="Reset" />
```

```
</ConstraintLayout>
```

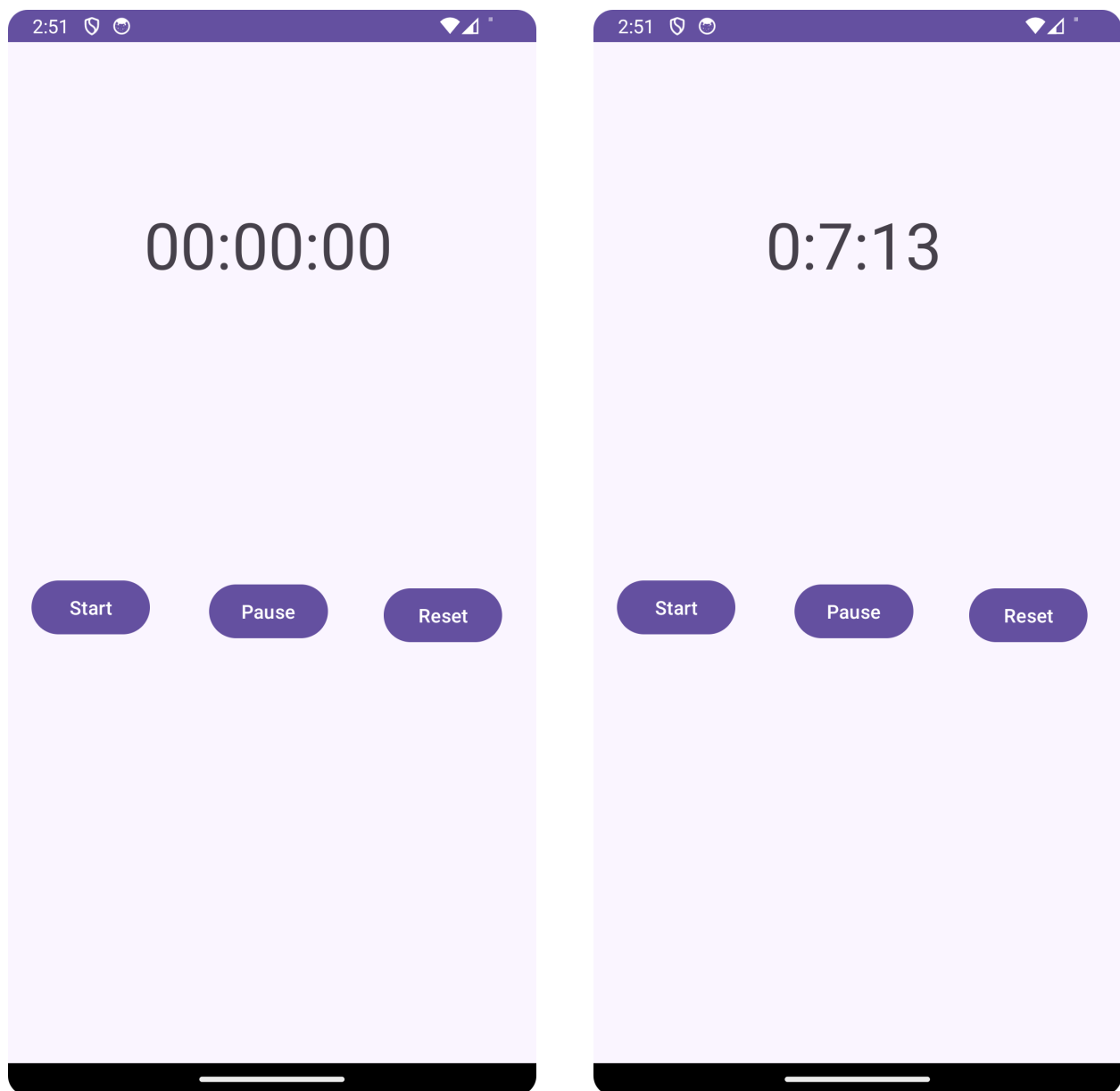
```
public class MainActivity extends AppCompatActivity implements Chronometer.C
    Chronometer c;
    int s=0,m=0,h=0;
    TextView tv;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        tv = findViewById(R.id.textView);
        c = findViewById(R.id.c);
        c.setBase(0);
```



```

        c.setOnChronometerTickListener(this);
    }
    @Override
    public void onChronometerTick(Chronometer chronometer) {
        s++;
        if(s==60)
        {
            m=m+1;
            s=0;
        }
        if(m==60)
        {
            h+=1;
            m=0;
        }
        tv.setText(h+":"+m+":"+s);
    }
    public void Start(View v) {
        c.start();
    }
    public void Pause(View v) {
        c.stop();
    }
    public void Reset(View v) {
        c.stop();
        s=0;
        tv.setText("00:00:00");
    }
}

```



17 Implement Multiplication Table

```
<LinearLayout>
```

```
<TextView
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
```

```
    android:id="@+id/tv"
```

```

        android:text="Enter the number : " />

<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/et"
    android:ems="10"
    android:inputType="textPersonName" />

<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/b1"
    android:onClick="generate"
    android:text="Generate" />

<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/et1"
    android:ems="10"
    android:gravity="start"
    android:inputType="textMultiLine" />
</LinearLayout>

```

```

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.linear);
    }

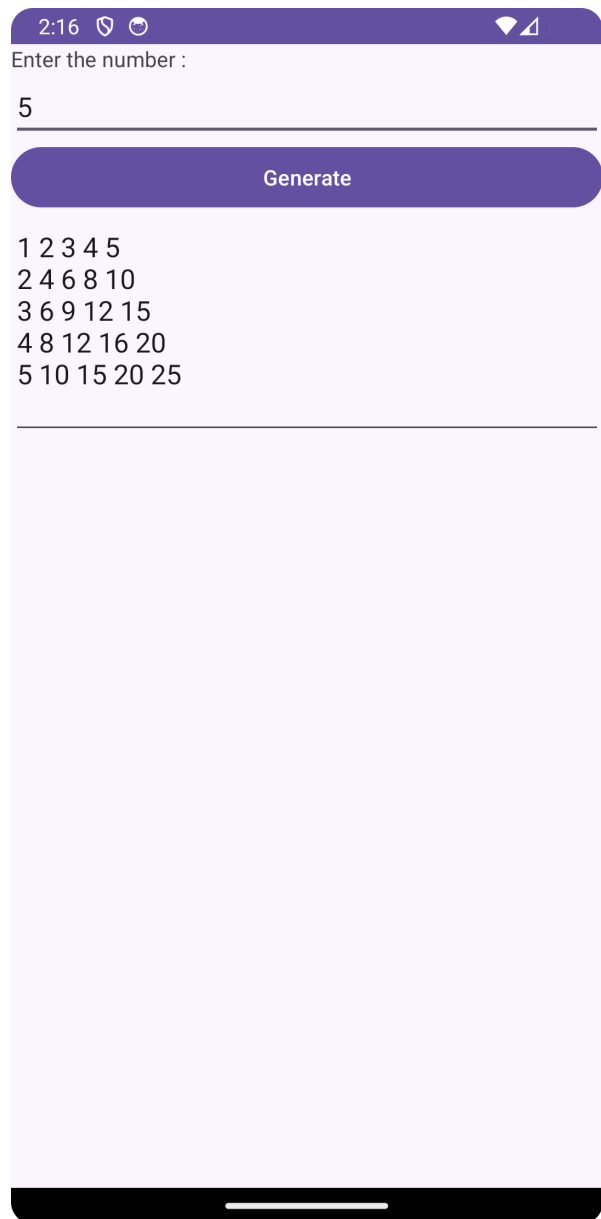
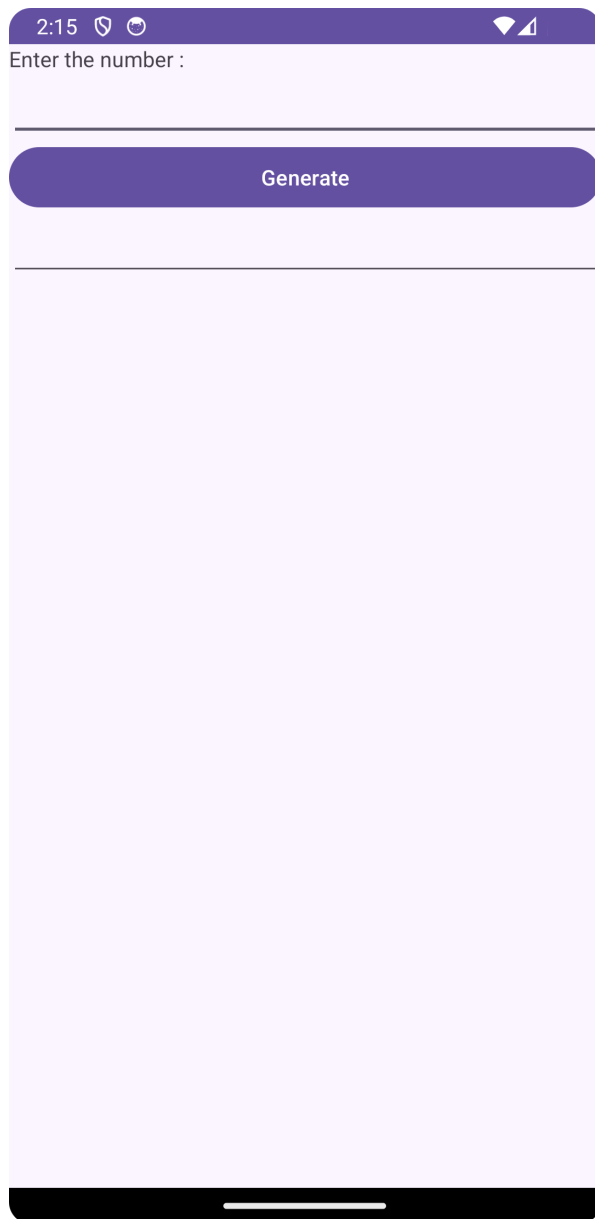
    public void generate(View v){

```

```

EditText et = findViewById(R.id.et);
EditText et2 = findViewById(R.id.et1);
String s1 = et.getText().toString();
int n = Integer.parseInt(s1);
int m = 1;
int i = 0;
int j = 0;
String s;
StringBuilder sb = new StringBuilder();
for (i = 1; i <= n; i++) {
    s = "";
    for (j = 1; j <= n; j++) {
        s = s + i * j + " ";
    }
    sb.append(s);
    sb.append("\n");
    et2.setText(sb.toString());
}
}
}

```

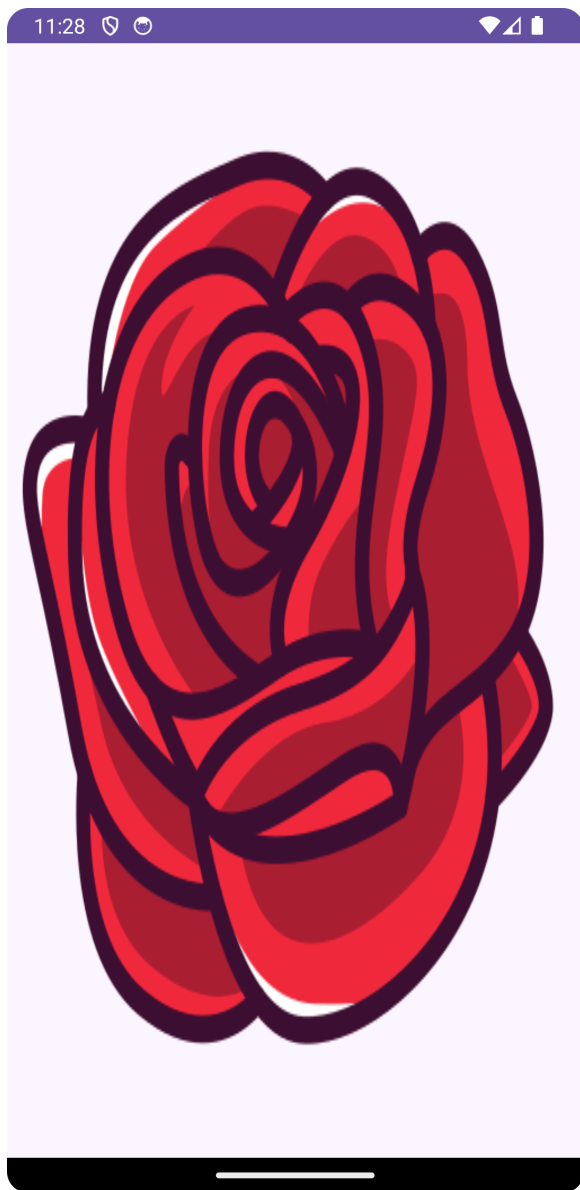


18 Change Background Image

```
<LinearLayout  
    android:orientation="vertical"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:background="@drawable/rose">
```

```
</LinearLayout>
```

```
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.linear);  
    }  
}
```



19 Change Background Color

```
<ConstraintLayout  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:background="#C13232">
```

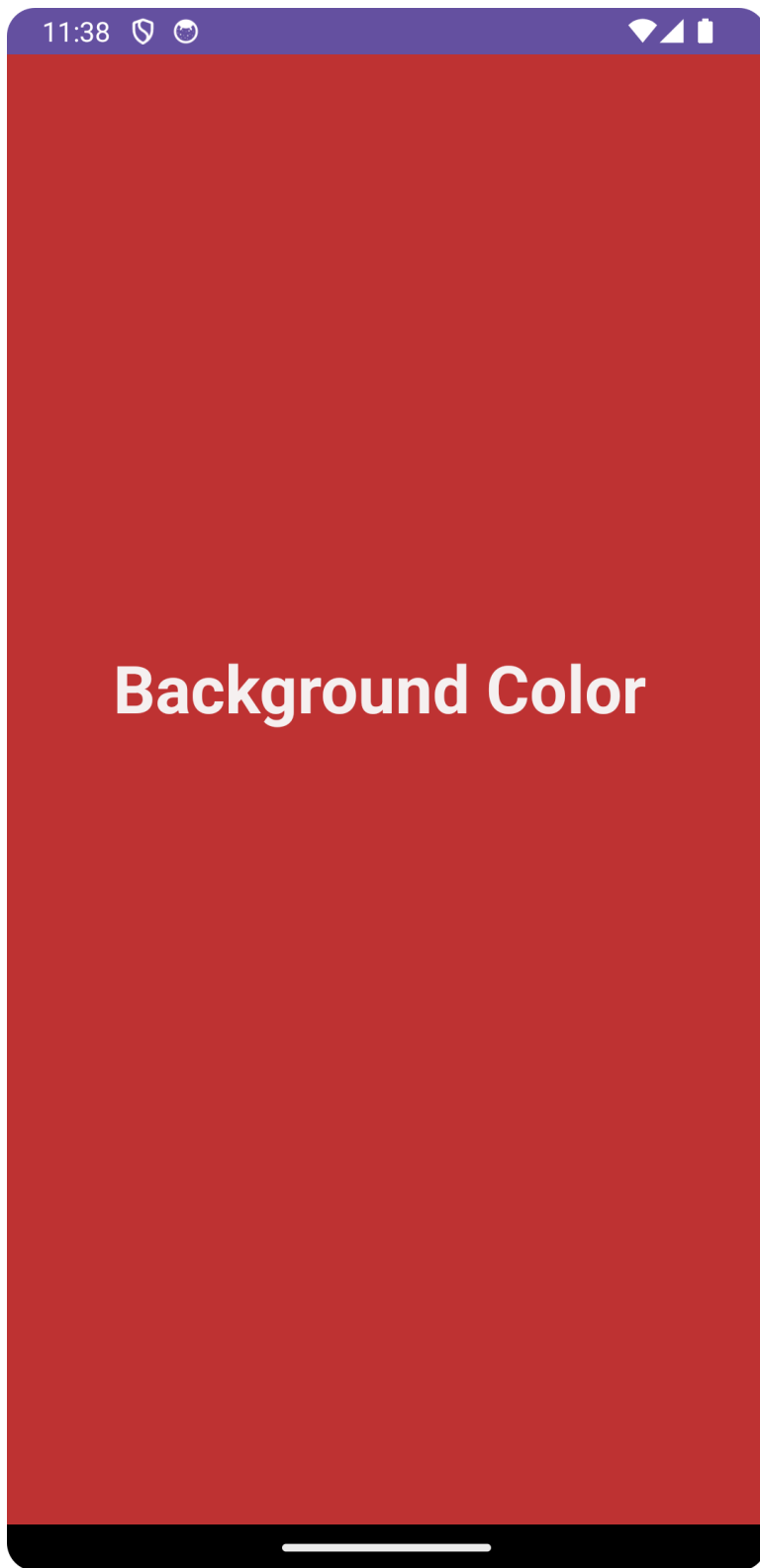
```
<TextView  
    android:layout_width="297dp"  
    android:layout_height="123dp"
```

```
android:text="Background Color"
android:textColor="#F8F0F0"
android:textSize="34sp"
android:textStyle="bold"/>
```

```
</ConstraintLayout>
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

20 Implement Login

```
<LinearLayout>
```

```
<TextView
```

```
    android:id="@+id/textview"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Username"/>
```

```
<EditText
```

```
    android:id="@+id/editText"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="textPersonName"  
    />
```

```
<TextView
```

```
    android:id="@+id/textView"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:text="Password"/>
```

```
<EditText
```

```
    android:id="@+id/editText2"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:ems="10"  
    android:inputType="textPersonName"  
    />
```

```
<Button
```

```
    android:id="@+id/button"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"
```

```

        android:text="Register"
        android:ems="10"
        android:onClick="register"/>

<Button
    android:id="@+id/button2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"
    android:ems="10"
    android:onClick="login"
/>

<Button
    android:id="@+id/button3"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Reset"
    android:ems="10"
    android:onClick="reset"/>

</LinearLayout>

```

```

public class MainActivity extends AppCompatActivity {
    EditText e1,e2;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.linear);
        e1=findViewById(R.id.editText);
        e2=findViewById(R.id.editText2);
    }
    public void register(View v)
    {
        SharedPreferences sp = getSharedPreferences("My pref",0);

```

```

        SharedPreferences.Editor ed=sp.edit();
        ed.putString("Username",e1.getText().toString());
        ed.putString("Password",e2.getText().toString());
        ed.commit();
    }
    public void login(View v)
    {
        SharedPreferences sp = getSharedPreferences("My pref",0);
        String u = e1.getText().toString();
        String p = e2.getText().toString();
        String k = sp.getString("Username" ," ");
        String k2 = sp.getString("Password" ," ");
        if(k.equals(u)&& k2.equals(p)){
            Toast.makeText(this,"Login Success",Toast.LENGTH_SHORT).show();
        }
        else {
            Toast.makeText(this,"Login Failed",Toast.LENGTH_SHORT).show();
        }
    }
    public void reset(View v)
    {
        e1.setText("");
        e2.setText(" ");
    }
}

```

2:48

Username

Password

Register

Login

Reset

2:51

Username

babu

Password

123

Register

Login

Reset

Login Success